THE RELATIVE MOR FUNCTOR IS STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. The phrase 'follows immediately from the definitions' is retained verbatim; the definitional unpacking is recorded in the roadmap as the reviewer's checklist, and the sheaf step carries the real content.

Allowed standard prerequisites (editorial).

- Sheaf property of Mor. lemma-hom-functor-sheaf (same chapter, named). Provides that both sides are sheaves.

Source reference. Original lemma 05Y3: sources/stacks-criteria/pinned-original-source.tex, source lines 1517–1517. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $Z \to B$, $X \to B$, and $B' \to B$ be morphisms of algebraic spaces over S. Set $Z' = B' \times_B Z$ and $X' = B' \times_B X$. Then*

$$Mor_{B'}(Z', X') = B' \times_B Mor_B(Z, X)$$

in $Sh((Sch/S)_{fppf})$.

Proof. The equality as functors follows immediately from the definitions. The equality as sheaves follows from this because both sides are sheaves according to Lemma 05Y3 and the fact that a fibre product of sheaves is the same as the corresponding fibre product of pre-sheaves (i.e., functors). □

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.